Cinema Management System Use Case Descriptions

These replacement tables cover UC-10, UC-23, UC-13, UC-34, UC-07, UC-09 and UC-17 for Group 2. The original SRS is partly correct but incomplete; UC-09 describes a different function and requires replacement. The tables below align the descriptions with the current implementation and preserve the source field order, merged metadata rows, PRE/POST numbering and numbered AF/EF flows.

Review date 03 November 2026 Code revision 4f69bc9

Review results

UC-10	Partly correct. Remove Address; use password hashing, the 8–32 limit and current-password verification for email changes. Specify default four-minute OTP, resend/cancel, five failed checks, optional DOB/gender, all-account phone uniqueness and session renewal.
UC-23	Partly correct. Add actual limits, optional end date/poster/formats, showtime protections and catalog sort/page behaviour. Archive means Hidden and can cancel all empty unfinished showtimes atomically; it is not interchangeable with Ended. Remove the nonexistent manager genre filter.
UC-13	Incomplete. Make hold/cutoff/seat limits configurable, add isolated seats, actual voucher/point calculations, zero-total and repeated-confirmation paths. Stop UC-13 at the internal payment page; stock is deducted only at payment completion.
UC-34	Mostly correct. Add search/pagination and used-genre rename. Deletion is blocked by any linked movie, including Hidden/Ended. Confirm deletion before the server check. The genre model has no Active category status.
UC-07	Partly correct. Use Visible reviews and actual paid/ended-screening eligibility. Add star filters, sorting, five-item pages, separate personal review and the protected route for Ended movies.
UC-09	Incorrect functional content. The title says Manage Notification, but the source table describes a seat map. Replace the actor, flows, conditions and rules with Manager announcement preview/send/history/recall. Personal inbox reading remains UC-18.
UC-17	Partly correct. Replace Watched-only eligibility and Published status with the actual Paid + Upcoming/Watched + ended-screening rule and Visible status. Add Hidden-review restrictions, draft preservation, deletion ownership and the Ended route.

Original creator/date metadata is retained. Priority and frequency are planning labels, not runtime properties; UC-34 frequency is completed as Low and UC-09 frequency is revised to As needed for the implemented Manager workflow.

UC 10 Manage Profile

UC ID and Name:	UC-10: Manage Profile		
Created By:	DuongND	**Date Created:**	19/9/26
Primary Actor:	Customer	**Secondary Actors:**	Mail Service
Trigger:	Customer opens My Profile from the account menu.		
Description:	Allows an authenticated customer to view and update full name, phone number, date of birth and gender, change password, and change the registered email through current-password verification and an email verification code.		
Preconditions:	PRE-01: Customer is logged in with an Active Customer account. PRE-02: The session credentials remain valid against the current account record.		
Postconditions:	POST-01: On a successful profile update, the submitted profile fields are saved and the current session is refreshed. POST-02: On a successful password change, a password hash is saved and the current session ID is renewed. Other sessions using the old password become invalid on their next protected request. POST-03: The email changes only after successful code verification; the pending change is cleared and the current session is refreshed. POST-04: Successful profile, password and email changes generate activity log entries. Rejected validation requests do not save the requested change.		
Normal Flow	1. Customer opens My Profile. 2. System displays the stored full name, phone number, email, date of birth and gender, together with separate password-change and email-change forms. 3. Customer edits full name, phone number and, optionally, date of birth and gender, then selects Save. 4. System normalizes the name and validates the submitted fields: - Full name is required, has at most 50 characters, and contains letters and separating spaces. - Phone number is required, contains exactly 10 digits, and starts with 0. - A supplied date of birth is a valid date from 01/01/1900 to yesterday. - A supplied gender is Male, Female or Other.		

UC ID and Name:	UC-10: Manage Profile
	5. System verifies that the phone number is not used by any other account. 6. System saves the profile fields, refreshes the current session and records the profile update. 7. System displays a success notification and the updated profile.
Alternative Flows:	**AF1: Change Password** 1. From Step 2, Customer enters the current password, a new password and its confirmation. 2. System verifies the current password and validates that the new password has 8–32 characters, includes uppercase and lowercase letters and a digit, differs from the current password, and matches its confirmation. 3. System hashes and saves the new password, renews the current session ID, refreshes the session user and records the change. 4. System displays a password-change success notification. The current session remains usable. **AF2: Change Email Address** 1. From Step 2, Customer enters a new email and the current password, then requests a verification code. 2. System verifies the password and checks that the normalized email is valid, has at most 100 characters, differs from the current email and is unused by another account. 3. System creates a six-digit code, invalidates previous codes for that destination and requests email delivery to the new address. Code validity uses mail.otp.expiryMinutes, currently 4 minutes by default. 4. Customer enters the received code and selects Verify. 5. System verifies the pending request, code validity and email uniqueness again. 6. System saves the new email, marks the code as used, clears the pending request, refreshes the session and records the change. System requests an email-change notice to the old address. 7. System displays an email-change success notification. **AF3: Resend or Cancel an Email Change** 1. During AF2, Customer may request another code after the 60-second resend cooldown. System replaces the old code and resets the failed-attempt counter for that pending request.

UC ID and Name:	UC-10: Manage Profile
	2. Alternatively, Customer cancels. System clears the pending request; the registered email remains the same.
Exceptions:	**EF1: Invalid Profile Information or Duplicate Phone** 1. At Steps 4–5, System detects a missing or invalid name/phone, an out-of-range date, an unsupported gender or a duplicate phone. 2. System preserves the submitted profile values, displays field errors and returns to Step 3 without saving them. **EF2: Invalid Password Change** 1. In AF1, the current password is wrong, the new password violates the policy or confirmation differs. 2. System rejects the change and displays the corresponding errors for correction. **EF3: Invalid Email or Current Password** 1. In AF2, the email is invalid, unchanged or already used, or the current password is incorrect. 2. System displays the relevant errors and does not start the requested email change. **EF4: Missing Pending Request or Invalid Verification Code** 1. System rejects an incorrect, expired, used or malformed code. After five failed code checks, System clears the pending request and requires Customer to start again. 2. A missing pending request, or an email taken by another account while verification was pending, also ends that pending change. Customer may restart AF2. 3. A resend request inside the cooldown is rejected with the remaining wait time. **EF5: Invalid Session or System Failure** 1. An expired session requires login; a suspended/deleted account or changed credentials invalidate the old session. An invalid request verification token is rejected. 2. A database error produces a system-error response. If the code email does not arrive, Customer may resend after the cooldown; delivery is not guaranteed by the success of the request alone.
Priority:	Medium

UC ID and Name:	UC-10: Manage Profile
Frequency of Use:	Low
Business Rules:	• BR-PROF-01: Email is read-only in the general profile form; changing it requires the separate password-and-code flow. • BR-PROF-02: Phone uniqueness applies to all other accounts, irrespective of account status. • BR-PROF-03: Password policy is 8–32 characters with uppercase, lowercase and a digit; the new password must differ from the current one. • BR-PROF-04: The current password is required for both password changes and email-change requests. • Date of birth and gender are optional; blank values may be saved as unspecified. • Verification codes are single use and time limited. The dedicated resend action has a 60-second cooldown and a pending request permits at most five failed code checks.
Other Information:	Address is not a profile field in the current implementation. The email can be changed through AF2; it is not permanently immutable. Security and account events are logged without storing plaintext passwords.
Assumptions:	Profile and email-verification database structures are available. The configured mail service can deliver to the new address, and Customer can access that mailbox.

UC 23 Manage Movies

UC ID and Name:	UC-23: Manage Movies		
Created By:	DuongND	**Date Created:**	19/9/26
Primary Actor:	Manager	**Secondary Actors:**	None
Trigger:	Manager opens Movie Catalog from the management dashboard.		
Description:	Allows Manager to list, search, create, edit and archive movie records, maintain genre associations, supported formats, posters and trailers, and manage movie lifecycle information.		

UC ID and Name:	UC-23: Manage Movies
Preconditions:	PRE-01: Manager is authenticated with an Active Manager account. PRE-02: For creation or saving a movie, at least one selectable genre exists. Browsing the catalog does not require a genre to exist.
Postconditions:	POST-01: On successful creation or editing, the movie and its genre associations are saved together. POST-02: A valid replacement poster is stored when supplied; an unused former system-stored poster is cleaned up. POST-03: On successful archiving, the movie becomes Hidden; eligible unfinished showtimes are cancelled and their temporary holds are released. Historical movie and sales records remain. POST-04: Successful catalog changes are logged and confirmed to Manager.
Normal Flow	1. Manager opens Movie Catalog. 2. System displays movie records, status counts, search, status filters, sorting and pagination, with eight records per page. 3. Manager selects Add Movie or Edit for an existing movie. 4. System displays the form, populated with current values for an edit, and the available genres. 5. Manager enters or modifies title, synopsis, director, cast, duration, age rating, release date, optional end date, country, language, distributor, poster, YouTube trailer, supported formats and status; Manager selects at least one genre. 6. Manager submits the form. System checks required values, lengths, title-and-release-year uniqueness, dates, supported options, poster validity and compatibility with existing unfinished showtimes. 7. System saves the movie and replaces its genre associations in one transaction. 8. System records the change, returns to the movie list and displays a success notification.
Alternative Flows:	**AF1: Search, Filter and Sort** 1. From Step 2, Manager enters a title keyword, selects Now Showing, Coming Soon, Ended or Hidden, or changes sorting. 2. System displays matching records and counts, sorted by newest record, title A–Z or release date; Manager may change pages. **AF2: Archive a Movie** 1. Manager selects Archive and confirms the action.

UC ID and Name:	UC-23: Manage Movies
	2. System checks the movie and every unfinished, non-cancelled showtime. 3. If none of these showtimes has an active ticket linked to a Paid or Pending booking, System cancels those showtimes, deletes their temporary holds and sets the movie to Hidden in one transaction. 4. System logs the operation and displays success, including the number of showtimes cancelled when applicable. A movie already Hidden causes no further cancellations. **AF3: Retain, Replace or Remove a Poster** 1. While editing, Manager keeps the current poster, uploads a replacement, or clears the current poster. 2. System validates a new file and saves the chosen result. A previous poster managed by the system is removed when no longer referenced by that movie. **AF4: Change Lifecycle Status** 1. Manager selects a valid status compatible with the release/end dates and showtime restrictions, then continues at Step 6. 2. Separately, JOB-03 promotes Coming Soon movies on reaching their release date and ends Now Showing movies after their recorded end date.
Exceptions:	**EF1: Invalid or Duplicate Movie Information** 1. At Step 6, required data is missing, a limit is exceeded, a genre does not exist, a format or age rating is unsupported, or the title already exists in the same release year. 2. System retains the form, displays field errors and does not save the movie. Manager corrects the form and resubmits. **EF2: Inconsistent Dates or Status** 1. A date is invalid, the end date precedes release, or the chosen status conflicts with the screening dates. 2. System rejects the affected values and displays the corresponding errors. **EF3: Conflict with Existing Showtimes** 1. An edit changes duration while unfinished showtimes exist, places unfinished showtimes outside the new screening dates, removes a format used by their rooms, or changes the status to Ended/Hidden while unfinished showtimes remain. 2. System rejects the conflicting edit. Manager must resolve the affected showtimes first or use AF2 when its conditions allow archiving.

UC ID and Name:	UC-23: Manage Movies
	3. In AF2, an active ticket on any unfinished showtime blocks the whole archive operation; no subset of the showtimes is cancelled. **EF4: Invalid Poster or Trailer** 1. The poster exceeds 5 MiB, is not an identifiable JPEG, PNG or WebP file, cannot be stored, or the trailer is not an accepted YouTube URL of at most 500 characters. 2. System displays the relevant error without saving the movie. A total upload request above 10 MiB is rejected. **EF5: Movie Missing or Request Fails** 1. The target movie no longer exists; System reports that it cannot be found. 2. An invalid session or request verification token prevents modification. A database failure produces a system-error response and rolls back a failed movie/genre or archive transaction.
Priority:	High
Frequency of Use:	Medium
Business Rules:	• BR-MOV-01: A trimmed title is required, at most 200 characters, and unique with its release year. • BR-MOV-02: Duration is an integer from 1 to 600 minutes; it cannot change while unfinished showtimes exist. • GB-03: Each movie has one supported age rating: P, C13, C16 or C18. These are application labels. • Release date is required, from 01/01/1900 through five years from today. End date is optional and cannot precede release or exclude an unfinished showtime. • Coming Soon requires a future release date. Now Showing/Ended cannot precede release; Now Showing cannot be after the end date. Hidden is independent of the date range. • At least one existing genre is required. Any supplied format must be 2D, 3D or IMAX; formats used by unfinished showtime rooms cannot be removed. • Optional text limits: synopsis 4,000; director 100; cast 500; country 100; language 100; distributor 200; trailer URL 500 characters. • GB-11: Public catalog visibility is limited to Now Showing and Coming Soon. • Archiving sets Hidden, never physically deletes the movie. An unfinished showtime means its end time is in the future and it is not Cancelled.

UC ID and Name:	UC-23: Manage Movies
	• JOB-03: Release/end-date synchronization is automatic; an absent end date does not trigger automatic ending.
Other Information:	Genre assignment belongs here; genre creation and renaming belong to UC-34. Manager catalog filters cover title and status; the current management screen does not provide a genre filter. Archive and setting Ended in the edit form are different actions.
Assumptions:	Genre data, poster storage and movie/showtime database structures are available. JOB-03 runs when automatic lifecycle synchronization is expected.

UC 13 Book Ticket Online

UC ID and Name:	UC-13: Book Ticket Online		
Created By:		**Date Created:**	19/9/26
Primary Actor:	Customer	**Secondary Actors:**	None within UC-13; Payment Gateway is used by UC-14.
Trigger:	Customer selects a showtime and chooses Book Ticket or Continue from seat selection.		
Description:	Allows Customer to select and temporarily hold seats, choose optional concessions, apply one voucher and loyalty points, review a server-calculated total and create a Pending online booking for UC-14 payment.		
Preconditions:	PRE-01: Customer is authenticated with an Active Customer account before holding seats. The seat-selection page itself may be viewed without login. PRE-02: Showtime is Active, has not started and is before the configured online-sale cutoff. PRE-03: Movie is publicly available and the room is not under maintenance; valid seats and prices are configured.		
Postconditions:	POST-01: After a successful hold, the selected seats are reserved for the configured hold interval and checkout displays the remaining time.		

UC ID and Name:	UC-13: Book Ticket Online
	POST-02: On confirmation, one Pending Online booking, ticket records and any concession lines are saved with the validated amounts. Its hold keeps the original expiry time. POST-03: Customer reaches the internal payment page for UC-14. The booking is not yet Paid; concession stock has not yet been deducted and redeemable ticket QR codes have not yet been finalized. POST-04: Cancelling an unconfirmed checkout releases its holds. An expired or invalid reservation cannot be confirmed.
Normal Flow	1. Customer selects a movie showtime. 2. System displays the seat layout, Standard/VIP/Sweetbox prices, current seat availability and the maximum selectable seats. 3. Customer selects available seats and confirms the age declaration for a C13, C16 or C18 movie. 4. Customer selects Continue. If necessary, Customer logs in before the protected reservation step. 5. System checks showtime availability, sale cutoff, seat membership, quantity, current availability, maintenance and the isolated-seat rule, then holds the seats for the configured interval, default 10 minutes. 6. System displays checkout with selected seats, the hold countdown, ticket subtotal, purchasable concessions, usable wallet vouchers and available loyalty points. 7. Customer optionally chooses concessions, one voucher and/or points. 8. System validates the choices, recalculates the price on the server and displays ticket/concession subtotals, voucher discount, point discount and final amount. 9. Customer confirms the order before the reservation or online-sale window ends. 10. System rechecks the live hold, sale eligibility, concession stock, voucher eligibility and available points; System creates the Pending booking and its ticket/concession records in one transaction. 11. System links the existing holds to the booking without extending their expiry, records booking creation and opens the internal payment page. UC-14 handles settlement.
Alternative Flows:	**AF1: Cancel or Select Seats Again** 1. Before a booking is created, Customer cancels checkout. System releases its held seats, clears checkout and returns to seat selection.

UC ID and Name:	UC-13: Book Ticket Online
	2. If Customer successfully starts another seat selection in the same browser, System replaces the previous unconfirmed holds. **AF2: Change Concessions or Discounts** 1. At Step 7, Customer changes a quantity, voucher or point amount; System recalculates checkout and retains valid choices. 2. If an item is no longer sold or lacks the requested stock, System reports the problem and removes the invalid item choice from the recalculated checkout. Other valid choices and unexpired seat holds remain. 3. With JavaScript unavailable, Customer may select Update to request recalculation. **AF3: Voucher and Points Together** 1. Customer selects at most one eligible voucher and requests an allowed number of points. 2. System checks minimum order value against the gross ticket-plus-concession total. Voucher discount applies only to ticket subtotal. 3. System applies the point discount to the remaining whole bill after the voucher, without allowing a negative final amount. **AF4: Zero Amount or Repeated Confirmation** 1. A booking with final amount zero still proceeds to the payment page, where UC-14 offers direct free-booking confirmation without a gateway charge. 2. If Customer confirms the same checkout again after its booking exists, System opens the existing booking payment page instead of creating another booking.
Exceptions:	**EF1: Invalid Selection, Busy Seats or Isolated Seat** 1. At Step 5, no seats are selected, the configured limit is exceeded, a seat is outside the room, another sale/hold occupies it, it is under maintenance, the age declaration is missing, or the choice creates a prohibited isolated seat. 2. System rejects the hold and displays the specific reason. Customer reviews seat selection; a lock timeout asks Customer to retry. **EF2: Hold Expired or Sale Closed** 1. At Steps 5, 8 or 10, the hold has expired, a held seat is no longer usable, or the showtime is no longer available for online sale.

UC ID and Name:	UC-13: Book Ticket Online
	2. System prevents confirmation, ends the invalid checkout and returns Customer to seat selection or the movie list with an explanation. Expired holds/Pending bookings are cleaned up by expiry processing. **EF3: Invalid Voucher** 1. The voucher is missing/inactive, not yet valid, expired, assigned to another customer/movie, below its minimum order, or over a total/per-customer usage limit. 2. During recalculation, System removes the invalid voucher discount, displays an error and recalculates valid choices. At confirmation, any such error prevents booking creation. **EF4: Invalid Points or Concessions** 1. Points are not a nonnegative integer, exceed the available balance or exceed the remaining bill; or a concession quantity is invalid, exceeds 10 per item or exceeds stock. 2. System reports the affected input and recalculates valid choices. Confirmation is blocked until all choices pass validation. **EF5: Session, Request or Database Failure** 1. A missing/invalid customer session requires login; an invalid request verification token is rejected. 2. A failed booking transaction rolls back its booking, ticket and concession writes. System reports a system error; a booking success notification is not issued.
Priority:	High
Frequency of Use:	Very High
Business Rules:	• GB-01: SEAT_HOLD_MINUTES is configurable from 1–60, default 10. Confirming a booking does not restart the countdown. • BOOKING_MAX_SEATS is configurable from 1–20, default 8. The limit counts selected seat records; one Sweetbox counts as one selection and accommodates two people. • ONLINE_SALE_CUTOFF_MINUTES is configurable from 0–180, default 15. At or after start time minus that value, online holds and payments are closed.

UC ID and Name:	UC-13: Book Ticket Online
	• GB-02: Available, held/Pending, Sold and Maintenance are availability states; only currently available and usable seats can be selected. Creating a Pending booking does not mean payment has succeeded. • GB-03: C13/C16/C18 require explicit age confirmation; this flow does not calculate age from the profile birth date. • Standard price equals showtime base price; VIP adds the configured surcharge; Sweetbox costs two Standard prices and produces one ticket for that seat record. • The isolated-seat rule rejects a new one-seat gap in an available contiguous run, except when the run would have exactly one unselected record remaining; a lone Sweetbox is not an isolated single seat. • GB-05: At most one voucher per booking. Ownership, time window, movie scope and usage limits apply; a live Pending booking counts toward relevant usage reservations. • GB-04: Point value follows POINT_EXCHANGE_RATE, default 1,000 VND. Available points exclude points reserved by other live Pending bookings. Final amount = ticket total + concession total − voucher discount − point discount. • GB-10: Concessions are stock-checked at checkout and confirmation, but stock deduction and point settlement/earning occur only when UC-14 completes payment.
Other Information:	UC-13 ends at the internal payment page. Gateway processing, payment callbacks, Paid status, finalized QR codes and confirmation email belong to UC-14. A temporary hold does not reserve concession inventory.
Assumptions:	Room layouts, showtimes, prices and system configuration are available. Expiry cleanup runs. Customer can complete UC-14 before both the hold deadline and online-sale cutoff.

UC 34 Manage Movie Genres

UC ID and Name:	UC-34: Manage Movie Genres		
Created By:	Duongnd	**Date Created:**	19/9/26
Primary Actor:	Manager	**Secondary Actors:**	None

UC ID and Name:	UC-34: Manage Movie Genres
Trigger:	Manager opens the movie genre catalog from the management dashboard.
Description:	Allows Manager to create, rename, search and delete unused genres while preserving unique names and movie associations.
Preconditions:	PRE-01: Manager is authenticated with an Active Manager account. PRE-02: For editing or deletion, the selected genre exists.
Postconditions:	POST-01: A successful save creates a genre or updates its name without changing its identifier or movie links. POST-02: A successful deletion removes an unused genre; linked movie records are not deleted. POST-03: System records successful changes and displays the updated list. Rejected requests leave genre data unchanged.
Normal Flow	1. Manager opens Genre Management. 2. System displays genres in name order, their associated movie counts, search and pagination, with 10 genres per page. 3. Manager selects Add Genre or Edit for a genre. 4. System displays an empty name form or the existing name. 5. Manager enters the genre name and selects Save. 6. System trims the name and checks that it contains 1–50 characters and is unique without regard to letter case, excluding the genre being edited. 7. System creates the genre or updates its name. 8. System logs the change, returns to the list with the current search/page parameters and displays success.
Alternative Flows:	**AF1: Delete an Unused Genre** 1. From Step 2, Manager selects Delete and confirms the action in the interface. 2. System checks that the genre still exists and is not associated with any movie. 3. System deletes the unused genre, records the action and displays success in the refreshed list. 4. If Manager cancels confirmation, no deletion request is performed. **AF2: Search and Change Pages** 1. Manager enters a genre-name keyword or selects another page. 2. System displays matching genres and their movie counts. No matches produces the empty-result message.

UC ID and Name:	UC-34: Manage Movie Genres
	AF3: Rename a Genre Already in Use 1. Manager edits the name of a genre that is assigned to movies. 2. After normal validation, System updates the name while retaining the same genre identifier and associations. Movie listings subsequently use the new name.
Exceptions:	**EF1: Empty, Overlong or Duplicate Name** 1. At Step 6, the trimmed name is empty, exceeds 50 characters or duplicates another genre without regard to case. 2. System retains the entered name, displays the relevant field error and returns to Step 5 without saving. **EF2: Genre Still Assigned to Movies** 1. In AF1, at least one movie references the genre, including Hidden or Ended movies. 2. System blocks deletion, displays the associated movie count and retains the genre. Manager must remove or replace those associations before retrying. **EF3: Genre Not Found** 1. The requested edit/deletion references a missing genre. 2. System reports that the genre cannot be found; no replacement record is created for the missing edit target. **EF4: Session, Request or Database Failure** 1. Missing authentication, insufficient role or an invalid request verification token prevents changes. 2. A database failure displays a system-error response. A concurrent duplicate-name violation is reported as a duplicate genre name.
Priority:	Medium
Frequency of Use:	Low
Business Rules:	• BR-GEN-01: A trimmed genre name is required, has 1–50 characters and is unique case-insensitively. • BR-GEN-03: A genre can be deleted only when no Movie_Genre association exists, irrespective of movie status. • Renaming a used genre is allowed; deleting it is not. Genre deletion is physical when unused.

UC ID and Name:	UC-34: Manage Movie Genres
	• Movies may have multiple genres. Adding/editing a movie must retain at least one valid genre through UC-23. • Successful add, rename and delete actions are recorded in catalog activity logs.
Other Information:	Genre is reference data for UC-23 and public movie search. The current genre model has no Active/Inactive status.
Assumptions:	Genre and movie–genre association structures are available. A predefined genre set is optional; Manager can create the first genre here.

UC 07 View Movie Reviews and Ratings

UC ID and Name:	UC-07: View Movie Reviews & Ratings		
Created By:	Duongn	**Date Created:**	19/9/26
Primary Actor:	Guest, Customer	**Secondary Actors:**	None
Trigger:	User opens the reviews section of movie details; an authenticated customer may also open the review page for an Ended movie from booking history.		
Description:	Displays visible customer reviews, average score, review count and 1–5-star distribution, with star filtering, sorting and pagination. A customer sees their own review separately and can see whether they are eligible to use UC-17.		
Preconditions:	PRE-01: For the public movie detail route, the movie exists and is Now Showing or Coming Soon. PRE-02: For the customer review route, Customer is logged in and the movie exists and is not Hidden; Ended movies can be viewed there.		
Postconditions:	POST-01: System displays rating statistics and the selected page of visible reviews without changing review records. POST-02: For Customer, System displays the personal review or eligibility guidance. Creating, editing or deleting a review is performed by UC-17.		
Normal Flow	1. User opens a public movie detail page and its Reviews section. 2. System verifies that the movie is publicly accessible.		

UC ID and Name:	UC-07: View Movie Reviews & Ratings
	3. System calculates the total count, average score rounded to one decimal place and distribution across 1–5 stars from Visible reviews only. 4. System retrieves Visible reviews, newest first by default, with five reviews per page, showing reviewer name, star rating, creation date and comment. 5. For an authenticated Customer, System retrieves the personal review separately and excludes it from the others list; System checks eligibility to write/update and the personal review status. 6. System displays the statistics, personal review or guidance, the review list and filter/sort/page controls.
Alternative Flows:	**AF1: No Visible Reviews** 1. At Steps 3–4, System finds no Visible reviews. 2. System shows a zero average/count, empty star distribution and a no-reviews message. **AF2: Filter or Sort Reviews** 1. User selects a star level from 1–5 or chooses newest, highest rating or lowest rating. 2. System updates the others list and pagination. The overall rating summary remains based on all Visible reviews, regardless of this filter. 3. An invalid star filter is ignored; an invalid sort falls back to newest. No matching reviews produces the empty-result message. **AF3: View an Ended Movie as Customer** 1. Customer opens the movie review page from booking history. 2. If the movie is Ended and not Hidden, System displays its rating statistics and reviews on the protected review page. 3. If the movie is still public, System opens the Reviews section of its public detail page instead. **AF4: Continue to Write or Manage a Review** 1. Guest is prompted to log in before writing. An ineligible Customer sees eligibility guidance. 2. An eligible Customer can continue to UC-17. A personal Hidden review is displayed to its owner with the moderation reason and unavailable edit/delete controls.
Exceptions:	**EF1: Public Movie Unavailable** 1. The public movie ID is invalid, missing, Ended or Hidden.

UC ID and Name:	UC-07: View Movie Reviews & Ratings
	2. System redirects to the public movie list with an unavailable-movie message. **EF2: Protected Review Page Unavailable** 1. Customer's movie identifier is invalid, the movie is missing/Hidden, or the customer session is invalid. 2. System rejects the invalid identifier, redirects missing/Hidden movie access to booking history with an error, or requires login as appropriate. **EF3: Review Retrieval Failure** 1. A database error prevents loading the movie or reviews. 2. System returns a system-error response; review data is not modified.
Priority:	Low
Frequency of Use:	Medium
Business Rules:	• GB-07: Public lists, totals and averages use Visible reviews only. Hidden reviews do not contribute to the public score. • Review eligibility requires at least one booking with Status Paid, a ticket with Status Upcoming or Watched, and a showtime for that movie whose EndTime is at or before the current database time. • Average = sum of visible star ratings divided by their count, rounded half up to one decimal; an empty set displays zero. • The personal review is separate from the paginated others list, but a Visible personal review still contributes to the overall summary. • GB-11 governs the public detail page. The authenticated review page additionally permits Ended movies that are not Hidden. • Viewing does not require an approval workflow; new valid reviews have status Visible, not Pending Approval.
Other Information:	UC-07 is read-only. It shares the review panel with UC-17; buttons that submit, update or delete enter UC-17. A completed paid screening is the implementation criterion, not proof of physical check-in.
Assumptions:	Movie and review records are available. The system/database clock and showtime end times are consistent for the eligibility check.

UC 09 Manage Notification

UC ID and Name:	UC-09: Manage Notification		
Created By:	Duongnd	**Date Created:**	19/9/26
Primary Actor:	Manager	**Secondary Actors:**	Recipients: Admin, other Managers, Staff and Customers, depending on the target group.
Trigger:	Manager opens Manage Notification from the management dashboard.		
Description:	Allows Manager to compose, preview and immediately send in-app announcements to a supported recipient group or one customer, inspect send/read history and recall sent announcements.		
Preconditions:	PRE-01: Sender is authenticated with an Active Manager account. PRE-02: For sending, at least one eligible Active recipient exists; for SingleCustomer, the phone identifies an Active Customer. PRE-03: For recall, the selected announcement exists and is still Sent.		
Postconditions:	POST-01: A successful send saves one Sent announcement, one inbox notification for each actual recipient and the actual recipient count in one transaction. POST-02: A successful recall sets Recalled, records who recalled it and when, and deletes its copies from all recipient inboxes in one transaction; the announcement history remains. POST-03: A successful send or recall generates a notification activity log. Preview and search do not send messages. POST-04: A validation failure or zero-recipient send does not create a completed send.		
Normal Flow	1. Manager opens Manage Notification. 2. System displays the compose form and shared announcement history, including sender, target, category, created time, read/recipient counts and Sent/Recalled status, with 10 records per page. 3. Manager chooses All, Staff, Customer or SingleCustomer. For SingleCustomer, Manager enters the customer phone number. 4. Manager selects an allowed category, enters a title and message, then selects Preview.		

UC ID and Name:	UC-09: Manage Notification
	5. System validates the sender, target/category combination and content; if needed, it looks up the customer and verifies role and Active status. System counts eligible Active recipients and excludes the sender. 6. System shows the content preview and recipient count. For SingleCustomer, it also shows the customer name and masked email. 7. Manager reviews the preview and selects Confirm Send. 8. System revalidates the submitted information and checks for an equivalent send by the same sender within the previous 60 seconds. 9. System saves the announcement and inbox copies for the recipients eligible at send time, records the actual count and commits only when the count is greater than zero. 10. System logs the send and displays success with the actual recipient count and updated history.
Alternative Flows:	**AF1: Revise the Preview** 1. At Step 6, Manager returns to edit, changes the content or recipient choices, then requests Preview again. 2. System validates and displays a new preview. No notifications are created until Confirm Send. **AF2: Search and Filter History** 1. Manager filters by target group or category, searches title keywords or changes page. 2. System displays the matching history, newest first. All Managers can view the history; it is not limited to the current sender. **AF3: Recall an Announcement** 1. Manager selects Recall on a Sent announcement and confirms the interface warning. 2. System verifies the announcement is still Sent, marks it Recalled and deletes all linked inbox copies, including copies already read, in one transaction. 3. System retains the historical announcement, sender, recipient count and recall metadata, records the recall and displays success. 4. If Manager cancels the confirmation, the announcement remains Sent. **AF4: Recipient Set Changes After Preview** 1. Accounts may become active/inactive or be created between Preview and Confirm Send.

UC ID and Name:	UC-09: Manage Notification
	2. System uses the recipients eligible at actual sending, not the earlier preview count, and displays that actual count.
Exceptions:	**EF1: Invalid Content, Target or Category** 1. At Step 5 or 8, target/category is unsupported, title is empty or over 150 characters, or message is empty or over 1,000 characters after trimming. 2. System retains the form and shows the relevant errors; nothing is sent. **EF2: Invalid Single Customer** 1. The supplied phone is missing, not exactly 10 digits starting with 0, not found, belongs to a non-Customer account or belongs to an inactive customer. 2. System displays the corresponding phone error and blocks preview/send. **EF3: No Eligible Recipients** 1. Preview finds no eligible recipient, or actual fan-out produces zero recipients. 2. System reports that no eligible recipients exist. A send in progress is rolled back without retaining its announcement or inbox copies. **EF4: Duplicate Send** 1. An equivalent Sent announcement by the same Manager exists within 60 seconds. 2. System rejects the repeated send and displays a duplicate warning. No second batch is created. **EF5: Recall Target Missing or Already Recalled** 1. In AF3, the selected announcement does not exist or is no longer Sent. 2. System reports that it cannot be found or has already been recalled; it does not perform another recall. **EF6: Authentication, Request or Database Failure** 1. A missing/inactive Manager session, insufficient role, invalid action or request verification token blocks the operation. 2. A database failure rolls back the send/recall transaction and returns a system-error response.
Priority:	High

UC ID and Name:	UC-09: Manage Notification
Frequency of Use:	As needed
Business Rules:	• All targets Active Admin, Manager, Staff and Customer accounts; Staff targets Active Staff and Manager accounts; Customer targets Active Customers; SingleCustomer targets exactly the identified Active Customer. Sender is always excluded. • Allowed categories: All → General/System; Staff → General/Internal/System; Customer or SingleCustomer → General/Movie/System. • General means general announcement; Movie means movie/showtime; Internal means internal staff notice; System means system/maintenance notice. Automated Review, Refund and Voucher categories are not selectable here. • Trimmed title length is 1–150; trimmed message length is 1–1,000 characters. SingleCustomer uses a 10-digit phone starting with 0. • Sending through the interface uses Preview then Confirm Send. Server validates again at sending; preview does not freeze recipients or create a draft announcement. • Duplicate detection within 60 seconds compares sender, title, message, target group and target customer; category is not part of that comparison in the current code. • Any Active Manager can recall any Sent announcement, including another Manager's; there is no recall time limit. Recall deletes inbox copies but preserves announcement history. • Recipients are fixed when sent; accounts created later do not receive old announcements. Delivery is in-app only.
Other Information:	This UC covers Manager announcement administration. Reading personal inboxes and marking messages read belong to UC-18 and determine history read counts. Scheduled sending, email/SMS delivery, attachments, sent-message editing and voucher campaigns are outside the implemented scope.
Assumptions:	Announcement and inbox database structures are deployed. Recipients use their role-specific inboxes; read counts reflect stored read markers, not proof that a person understood the message.

UC 17 Rate and Review Movie

UC ID and Name:	UC-17: Rate & Review Movie		
Created By:	Duongnd	**Date Created:**	19/9/26
Primary Actor:	Customer	**Secondary Actors:**	None
Trigger:	Customer selects Write Review or Rate Movie from movie details or booking history.		
Description:	Allows an eligible customer to submit a 1–5-star rating and required comment, update the single review for a movie or delete their own Visible review. Ended movies remain reviewable unless Hidden.		
Preconditions:	PRE-01: Customer is authenticated with an Active Customer account. PRE-02: For creating/updating, Customer has a Paid booking for the movie, with at least one ticket in Upcoming or Watched status and a showtime EndTime not later than the database time. PRE-03: For creating/updating, the movie exists and is not Hidden. An existing Hidden personal review cannot be edited or resubmitted. PRE-04: For deletion, the selected review belongs to Customer and is Visible.		
Postconditions:	POST-01: A successful submission creates one Visible review or updates the existing Visible review and its update time. POST-02: A successful deletion physically removes the personal Visible review. POST-03: Subsequent rating summaries reflect the current Visible review records. Successful create/update/delete actions are logged and confirmed. POST-04: Invalid submissions retain the draft rating/comment for redisplay; they do not save a changed review.		
Normal Flow	1. Customer opens the movie review interface from movie details or booking history. 2. System checks the movie and Customer’s completed paid screening eligibility, and retrieves any existing personal review. 3. If there is no existing review and Customer is eligible, System shows the rating/comment form. 4. Customer selects an integer rating from 1–5, enters a comment and selects Submit.		

UC ID and Name:	UC-17: Rate & Review Movie
	5. System rechecks movie availability, screening eligibility and whether a personal review already exists. 6. System validates the rating and checks that the trimmed comment contains 10–1,000 characters. 7. System inserts a review with status Visible and the current creation time. 8. System clears the submitted draft, records the action and redirects to the appropriate review interface. 9. System reloads the rating summary and lists from current Visible records and displays success.
Alternative Flows:	**AF1: Update an Existing Visible Review** 1. At Step 2, System finds Customer's Visible review and displays its current rating/comment. 2. Customer selects Edit, changes the values and submits. 3. System repeats eligibility and input validation, updates that review's rating/comment and update time, records the action and shows the updated summary. 4. A repeated create submission for the same customer/movie updates the existing Visible review rather than adding a second one. **AF2: Delete a Personal Visible Review** 1. Customer selects Delete on the personal review and confirms the interface prompt. 2. System checks the review identifier, ownership and Visible status, then deletes the review. 3. System logs the deletion, reloads statistics and displays success. If Customer cancels, no deletion occurs. 4. Deletion itself checks ownership/Visible status; it does not rerun the completed-screening eligibility query. **AF3: Review an Ended Movie** 1. Customer opens the review action in booking history for an Ended movie. 2. System uses the protected review page, checks that the movie is not Hidden and applies the same eligibility and input rules. 3. For a movie still Now Showing/Coming Soon, System uses its public detail Reviews section instead.

UC ID and Name:	UC-17: Rate & Review Movie
	AF4: Submit Again After Deletion 1. After a successful deletion, Customer may create another review only if the current movie and completed-screening eligibility checks still pass. 2. System creates a new Visible review through the normal flow.
Exceptions:	**EF1: Customer Not Eligible** 1. At Step 2 or 5, Customer has no qualifying Paid booking/ticket for a screening that has ended. 2. System prevents create/update and displays the eligibility message. An unpaid, cancelled/exchanged ticket or a screening that has not ended does not qualify. **EF2: Invalid Rating or Comment** 1. At Step 6, the rating is outside 1–5 or not an accepted integer, or the trimmed comment is missing, shorter than 10 or longer than 1,000 characters. 2. System preserves the submitted rating/comment as a draft, displays the error on the relevant movie interface and lets Customer correct and submit again. **EF3: Personal Review Hidden by Manager** 1. System finds the existing personal review has status Hidden. 2. System displays its moderation reason to its owner and blocks editing, resubmission and customer deletion. Posting again cannot bypass the hidden status. **EF4: Movie or Review Unavailable** 1. Creating/updating targets a missing/Hidden movie, or deletion targets a missing review, another user's review or a non-Visible review. 2. System reports the unavailable movie/review and does not change it. For unavailable movie access, the protected page returns Customer to booking history; malformed identifiers are rejected. **EF5: Session, Request or Database Failure** 1. Missing/invalid authentication or an invalid request verification token prevents the operation. 2. A database failure returns a system-error response. A concurrent duplicate for the same customer/movie is handled as an update to the existing Visible review; a Hidden review remains locked.
Priority:	Low

UC ID and Name:	UC-17: Rate & Review Movie
Frequency of Use:	Medium
Business Rules:	• GB-07: Eligibility is Paid booking + Upcoming/Watched ticket + showtime ended, for that customer and movie. Physical check-in is not an extra condition in this implementation. • Maximum one review record per customer/movie. Updates overwrite the rating/comment on that record; a valid new review is Visible immediately, not Published or Pending Approval. • Rating is an integer 1–5. Comment is required and has 10–1,000 characters after trimming. • Hidden reviews do not contribute to public statistics and cannot be edited, resubmitted or deleted by Customer. Manager moderation is a separate function. • Customers can edit/delete only their own Visible review. Ended movies support writing through the protected route; Hidden movies do not support new/updated reviews. • Average, star distribution and review count are calculated from the current Visible set; no stored movie-average update is required.
Other Information:	UC-07 covers review viewing, filtering and sorting. A Hidden personal review may generate a moderation notification read through UC-18. Successful actions are logged; logging is not a review approval stage.
Assumptions:	Paid bookings, ticket states and showtime end times are accurate. Eligibility does not depend on an automatic job first converting every ticket to Watched.

Implementation traceability

Reference SRS: Group2_SRS (2).docx, supplied from D:/Download. Implementation paths below are relative to CinemaManagementSystem_SE2053_NET_Group2/SE2053-SWP391-G2. Java files are under src/main/java/vn/edu/fpt; JSP/tag/JS files under src/main/webapp; migrations under sql. The UC09 plan is at repository-root docs/plans. Source rules and SQL were checked against the executable Java/JSP paths; stale comments were not treated as implemented behaviour.

UC 10

GET/POST /customer/profile

ProfileServlet; ProfileService; EmailChangeService; FullNameRule; AuthFilter; UserDAO; EmailVerificationDAO

Data: User; EmailVerification; ActivityLog

SQL and related source: UC10_profile_birthdate_gender.sql; config.properties (mail.otp.expiryMinutes only)

UC 23

GET/POST /manager/movies

MovieManagementServlet; MovieService; MovieDAO; PosterStorage; MovieStatusSyncListener

Data: Movie; Genre; Movie_Genre; Showtime; Ticket; Booking; Seat_Hold; ActivityLog

SQL and related source: UC23_movie_lifecycle_details.sql; UC23_add_movie_formats.sql

UC 13

GET /booking; POST /customer/booking/hold; GET/POST /customer/booking/checkout; /customer/booking/payment (handoff)

BookingSeatServlet; BookingHoldServlet; BookingCheckoutServlet; BookingFlow; BookingService; BookingDAO; SeatMapService; SeatGapRule; TicketPricing; SystemConfigParam

Data: Seat; Seat_Hold; Showtime; Room; Movie; Booking; Ticket; BookingConcession; Concession; Voucher; User; SystemConfig

SQL and related source: UC13_book_ticket_online.sql; UC28_vouchers.sql; UC34_movie_scoped_vouchers.sql; UC33_system_config.sql

UC 34

GET/POST /manager/genres

GenreManagementServlet; GenreService; GenreDAO

Data: Genre; Movie_Genre; ActivityLog

SQL and related source: cinemaDB.sql

UC 07

GET /movies/detail?id=…; GET /customer/reviews?movieId=… for Ended movies

MovieDetailServlet; MovieReviewServlet; MovieService; ReviewService; ReviewDAO; ReviewSummary; review-panel.tag

Data: Review; User; Movie; Ticket; Booking; Showtime

SQL and related source: UC17_movie_review.sql

UC 09

GET/POST /manager/announcements

NotificationManagementServlet; AnnouncementService; AnnouncementDAO; AnnouncementForm; NotificationTarget; NotificationCategory; UserDAO; announcements.jsp

Data: Announcement; CustomerNotification; User; ActivityLog

SQL and related source: UC09_manage_notifications.sql; docs/plans/UC09-manage-notifications.md (context, checked against implementation)

UC 17

GET/POST /customer/reviews; /movies/detail#reviews (interface)

MovieReviewServlet; ReviewService; ReviewDAO; ReviewForm; MovieDetailServlet; review-panel.tag; movie-review.js

Data: Review; Movie; User; Booking; Ticket; Showtime; ActivityLog

SQL and related source: UC17_movie_review.sql

5. Background Job Inventory

The shift auto-close task belongs to UC-22 and currently has no separate JOB identifier.

5.1 Release Expired Temporary Seat Holds (JOB-01)

Trigger: Scheduled — first runs one minute after application startup, then repeats one minute after the previous execution finishes.

Purpose: Automatically cancels expired Pending online bookings and removes expired temporary seat holds so that eligible seats can become available for other customers. The hold duration follows `SEAT_HOLD_MINUTES`, with a default of 10 minutes.

Processing Steps:

1. Find Online bookings with Pending status that no longer have any unexpired seat holds associated with the booking.
2. Change the identified bookings from Pending to Cancelled and their Upcoming tickets to Cancelled.
3. Remove the seat holds associated with the cancelled bookings. Save the booking, ticket and associated hold changes within the cancellation transaction.
4. Remove any remaining temporary seat holds whose expiration time has passed, including holds for checkout sessions that have not created a booking.
5. Allow released seats to become available for selection, provided that they are not otherwise occupied or under maintenance.
6. Log the number of cancelled bookings and the number of remaining expired holds removed by the cleanup operation.

Failure behavior:

Scenario	System behavior
Database or processing error during booking cancellation	Roll back the cancellation transaction, log a warning and retry during the next scheduled execution. Remaining operations in the current execution are not continued.
Error while removing remaining expired holds	Log a warning and retry during the next scheduled execution. Previously committed booking cancellations remain effective.
No expired bookings or holds found	Complete without changing data. This is normal and expected.

Customer is still viewing an expired checkout	The job does not push a notification to the browser. The booking flow detects expiration when the customer continues or reloads checkout and displays the corresponding message.

5.2 Dispatch Showtime Reminder Emails (JOB-02)

Trigger: Scheduled — runs immediately after application startup, then repeats 15 minutes after the previous execution finishes.

Purpose: Sends reminder emails containing movie information, showtime, selected seats, ticket codes and available e-ticket QR codes to customers with qualifying paid bookings.

Processing Steps:

1. Find Paid bookings associated with a customer and at least one Upcoming ticket, where the showtime starts more than one hour and no more than two hours from the current database time.
2. Create one reminder queue record for each qualifying booking that does not already have a reminder record.
3. Select up to 50 eligible reminder records per execution. Each selected record must remain unprocessed, have fewer than three recorded attempts and have reached its next permitted attempt time.
4. Record the attempt and schedule the next permitted attempt at least 15 minutes later.
5. Retrieve the current customer email, movie information, showtime and Upcoming ticket details. Confirm that the booking remains Paid.
6. Send one personalized reminder email per booking through the configured email service.
7. On successful sending, record the sent time and clear the previous delivery error so that the reminder is not selected again.
8. On failed sending, record the delivery error and retain the reminder for another eligible attempt.
9. Log the number of reminder emails successfully sent during the execution.

The one-to-two-hour window applies when reminders are first queued. Retry selection does not recheck this window, so a retry may occur closer to the showtime.

Failure behavior:

Scenario	System behavior

Email service is unavailable, SMTP is not configured or delivery fails	Record the delivery error and log a warning. Retry after the next permitted attempt time, up to three total attempts, including the first attempt.
Maximum attempt count reached	Stop automatically selecting that reminder for further delivery attempts. Retain its processing information.
Queued booking no longer provides qualifying reminder details	Mark the queue record as processed without sending an email, preventing repeated selection of an obsolete reminder.
Database error during reminder processing	Log a warning and end the current execution. Eligible unprocessed records may be selected during a later execution.
Email was sent but saving its sent status fails	The reminder may be selected again because successful delivery was not recorded. The current implementation does not guarantee exactly-once email delivery.
No reminders are eligible for processing	Complete without sending emails. This is normal and expected.

5.3 Synchronize Movie Screening Statuses (JOB-03)

Trigger: Scheduled — runs once approximately one minute after application startup, then daily around midnight according to the application server's local time.

Purpose: Automatically updates movie screening statuses according to release dates, end dates and unfinished showtimes, keeping the movie catalog consistent with the screening lifecycle.

Processing Steps:

1. Determine the current date using the application server's local clock.
2. Find movies with Coming Soon status whose release date is on or before the current date.
3. Change the identified movies from Coming Soon to Now Showing.
4. Find movies with Now Showing status whose recorded end date is before the current date.
5. Check that each candidate has no unfinished, non-cancelled showtime. A showtime is unfinished when its end time is later than the current database time.
6. Change eligible movies from Now Showing to Ended. Movies with unfinished showtimes remain Now Showing for reconsideration during a later execution.

7. Leave Hidden movies unchanged. Movies without an end date are not automatically changed to Ended by this job.
8. Record successful status transitions and affected movie titles in the application log and catalog activity history, identifying the action as performed by the system.

Failure behavior:

Scenario	System behavior
Database or processing error	Log a warning and retry during the next scheduled execution. Status changes already committed before the error remain effective.
Application was stopped during a scheduled execution	The startup execution evaluates the current dates and processes eligible movies after the application starts again.
Movie has passed its end date but still has an unfinished, non-cancelled showtime	Keep the movie in Now Showing status and reconsider it during a later execution.
Movie has no end date or already has Hidden status	Do not automatically change it to Ended.
No movies require a status change	Complete without changing data or creating a status-change activity entry.

5.4 Automatically Close Expired Counter Shifts (UC-22)

Trigger: Scheduled — first runs one minute after application startup, then repeats one minute after the previous execution finishes.

Purpose: Automatically closes counter shifts that have reached the configured maximum duration, preventing forgotten open shifts and preserving their closing sales totals.

Processing Steps:

1. Read the configured shift duration from `SHIFT_HOURS`. The default duration is four hours; the management interface permits values from 1 to 12 hours.
2. Find open shifts whose start time plus the configured duration is on or before the current database time.
3. Process each identified shift in a separate transaction and verify that it is still open.

4. Calculate the shift totals, including ticket count, ticket revenue, concession revenue, payment totals by Cash/Card/Transfer and recorded refunds.
5. Save the closing totals and set the shift end time to its start time plus the configured duration.
6. Record the actual processing time separately as the closing timestamp and identify the closure as performed by the system.
7. Leave the counted cash amount unspecified. Staff or Manager records the actual cash count through the shift-management workflow.
8. Commit the shift closure and continue with the remaining eligible shifts.
9. When shifts have been closed successfully, log the number of automatically closed shifts and record a counter activity entry.

Failure behavior:

Scenario	System behavior
Database or processing error while closing a shift	Roll back that shift's transaction, log a warning and stop the current execution. Remaining eligible shifts are reconsidered during the next scheduled execution. Previously committed closures remain effective.
Shift was already closed by another operation	Skip the shift without closing it again or overwriting its closing totals.
No expired open shifts found	Complete without changing data. This is normal and expected.
An automatically closed shift has no cash count	Keep the cash count unspecified until an authorized user records it. Automatic closure does not invent a counted cash amount.

6. External API Inventory

The tables below follow the SRS fields URL, Purpose, Caller and Auth. Sections 6.1–6.3 describe the implemented VNPay HTTP interfaces; Sections 6.4–6.5 describe the SMTP and media-embed interfaces used by the project. Relative application paths are prefixed by the deployed application context.

6.1 VNPay Payment Redirect (API-01)

URL	GET: `payment.vnpay.payment-url`; example sandbox endpoint: `https://sandbox.vnpayment.vn/paymentv2/vpcpay.html`
Purpose	Opens VNPay checkout for an eligible Pending online booking with its exact payable amount and a unique payment-attempt reference.
Caller	Customer's browser, redirected by Cinema Management System after the authenticated customer starts payment in UC-14.
Auth	Yes — the application signs the request parameters with HMAC-SHA512 using the configured merchant secret and includes `vnp_TmnCode` and `vnp_SecureHash`. Starting payment inside the application requires an Active Customer session, booking ownership and a valid request verification token.

The application builds a VNPay 2.1.0 payment URL and redirects the browser; it does not submit a server-to-server payment-creation REST request. The amount is an integer VND value multiplied by 100 in vnp_Amount. Payment expiry is limited by the configured gateway duration, remaining seat-hold time and online-sale cutoff. Zero-value bookings are confirmed internally without contacting VNPay.

6.2 VNPay Webhook / IPN (API-02)

URL	GET / POST: `/api/v1/commerce/payment/webhook/vnpay`
Purpose	Receives a signed VNPay payment result and settles a matching eligible booking, including ticket QR issuance, voucher usage and loyalty updates.

Caller	VNPay payment gateway, through an incoming server-to-server IPN notification.
Auth	Yes — HMAC-SHA512 signature validation with the configured merchant secret; merchant terminal code, required parameter formats and optional currency/algorithm fields are also checked. No customer login session, API-key header or bearer token is required by this endpoint.

GET is the primary supported IPN method; POST is also accepted through request parameters. The callback URL is configured in the VNPay merchant environment, rather than included as an IPN parameter in the payment URL. A configured callback URL must use HTTPS.

Processing verifies the payment-attempt reference, transaction information, amount and booking eligibility. Return/IPN settlement is idempotent. A successful gateway result that cannot settle an expired or mismatched booking is recorded for review instead of issuing tickets.

The endpoint returns JSON fields RspCode and Message. Current application codes are 00 (accepted/handled), 01 (transaction not found or not accepted), 97 (invalid parameters/signature/callback validation), 02 (transaction-data validation failure) and 99 (database/system error). RspCode 00 acknowledges processing and does not by itself mean that the booking was marked Paid.

6.3 VNPay Browser Return (API-04)

URL	GET: `/customer/booking/payment/return`
Purpose	Verifies the signed VNPay browser return, attempts idempotent booking settlement and redirects the customer to the booking payment/result page.
Caller	Customer’s browser after VNPay redirects it to the application’s configured Return URL.
Auth	Yes — the current `/customer/*` authentication filter requires an Active Customer session before the servlet is reached. VNPay return parameters must independently pass HMAC-SHA512 verification and merchant/parameter validation. Result lookup also checks booking ownership; a browser visit alone is not proof of payment.

The Return flow uses the same signature verifier and booking-settlement service as IPN, so it can complete payment when public IPN delivery is unavailable. The existing authentication filter still applies if the browser session has expired. The configured Return URL requires HTTPS, except that the current implementation explicitly permits HTTP for localhost or 127.0.0.1 during local development.

6.4 SMTP Email Delivery (EXT-01)

URL	SMTP: configured `mail.smtp.host` and `mail.smtp.port`; defaults: `smtp.gmail.com:587`. This is an SMTP service interface, not an HTTP endpoint.
Purpose	Delivers registration and email-change verification codes, password-reset links, email-change notices, booking confirmations with e-tickets, and showtime reminders.
Caller	Cinema Management System server through EmailService using Jakarta Mail / Angus Mail; the external SMTP server receives and relays the messages.
Auth	Yes by default — configured sender email and sender password/application password through SMTP authentication. `mail.smtp.auth` and `mail.smtp.starttls.enable` default to true; the implementation configures TLSv1.2 for SMTP connections.

Most account and booking emails are submitted asynchronously. JOB-02 sends showtime reminders through a synchronous delivery method so its queue can record success or failure and apply its retry policy. Missing SMTP credentials do not provide real email delivery; selected development paths log content, while reminder delivery reports failure. No email REST API, SMS provider or inbound email-delivery webhook is implemented.

6.5 YouTube Trailer Embed (EXT-02)

URL	GET: `https://www.youtube-nocookie.com/embed/{videoId}`
Purpose	Displays the configured movie trailer in an embedded player on the public movie-detail page.

Caller	Visitor’s browser, which loads the external iframe generated by the movie-detail page.
Auth	No application-supplied API key, OAuth token or customer login is used for the embed. The application validates the configured YouTube trailer URL and extracts its 11-character video ID before rendering the iframe.

This is a browser-side media embed. The project does not call the YouTube Data API or import movie metadata from an external movie-catalog API.

6.6 Implementation Scope Notes

- API-03 in the original SRS refers to a SePay webhook. The current project declares SePay as a payment-method value, but has no SePay gateway or /api/v1/commerce/payment/webhook/sepay servlet. API-03 is therefore reserved and is not documented as an implemented endpoint; API-04 identifies the implemented VNPay Return interface.
- E-ticket QR images are generated locally using ZXing. Counter Cash/Card/Transfer entries record the staff-selected payment method; they do not establish a separate VietQR, card-acquiring or bank-payment API integration.
- Tailwind CSS and Google Fonts / Material Symbols are loaded as external presentation assets. They are frontend dependencies rather than business-service API endpoints.
- Movie suggestions, seat-status polling and checkout fetch requests call the application’s own routes. They are internal web interfaces and do not represent additional third-party APIs.

7. Use Cases (UC)

ID	Use Case	Feature	Use Case Description
07	View Movie Reviews & Ratings	Movie Browsing	Displays Visible movie reviews, average ratings and the 1–5-star distribution, with star filters, sorting and pagination. Shows a signed-in customer's own review separately and indicates whether the customer is eligible to write or update a review.
09	Manage Notification	Notification Management	Allows Managers to preview and send in-app announcements to eligible Active users by audience group or a single customer, search announcement history, view recipient/read counts, and recall Sent announcements while preserving their history.
10	Manage Profile	Profile Management	Allows authenticated customers to view and update their full name, phone number, optional date of birth and gender, change their password, and change their registered email through current-password verification and an email OTP.
13	Book Ticket Online	Online Booking	Enables customers to select and temporarily hold eligible seats, optionally add concessions and apply one voucher and/or available loyalty points, then confirm a Pending booking and proceed to the internal payment page.
17	Rate & Review Movie	Feedback & Interaction	Enables customers with a Paid booking and an Upcoming or Watched ticket for a screening that has ended to create or update one Visible review per movie with a 1–5-star rating and required comment, or delete their own Visible review.
23	Manage Movies	Movie Catalog	Allows Managers to search, filter and maintain the movie catalog, including movie details, genres, formats, posters, trailers, dates and statuses, and archive movies as Hidden while protecting unfinished showtimes with active tickets.
34	Manage Movie Genres	Movie Catalog	Allows Managers to search and paginate genres, create genres with unique names, rename existing genres even when linked to movies, and delete only genres that are not associated with any movie.

37	View Online Ticket Sales History	Sales History & Reporting	Allows authorized Managers and permitted Staff to view completed online orders by payment date, four summary figures and daily/monthly collections. Supports five quick periods, custom dates, order details, 10 orders per page and CSV export.

8. UC 37 View Online Ticket Sales History

UC ID and Name:	UC-37: View Online Ticket Sales History		
Created By:	Group 2	**Date Created:**	06/11/26
Primary Actor:	Manager	**Secondary Actors:**	Staff explicitly granted VIEW_ONLINE_REPORTS by Admin (UC-31)
Trigger:	The actor selects Reports & Revenue → Online Ticket Sales History ("Báo cáo & doanh thu" → "Lịch sử bán vé online") from the management sidebar.		
Description:	Allows an authorized actor to view completed online ticket orders, filter them by payment date, and review four summary figures together with a daily or monthly collections chart. The actor can browse the paginated paid-order history, view an order's ticket/concession details and export the matching orders to CSV.		
Preconditions:	PRE-01: The actor is logged in with an Active Manager or Staff account. PRE-02: The actor has VIEW_ONLINE_REPORTS permission. Manager has this permission by default; Staff must be granted it through UC-31.		
Postconditions:	POST-01: For valid filters, System displays the matching paid online orders, four summary figures and the collections chart, or returns the requested order details/CSV. POST-02: Summary figures and chart totals cover the entire selected period, independently of the displayed history page. POST-03: Viewing, filtering, paging and exporting do not change bookings, tickets or payment records. POST-04: Invalid requests or unavailable data produce the corresponding validation/access/system response; a data-loading failure is not presented as zero sales.		
Normal Flow	1. The actor opens Reports & Revenue and selects Online Ticket Sales History. 2. System checks authentication and permission, then opens the history with Today selected by default. 3. The actor keeps Today or selects another quick period, or enters From Date and To Date and selects View History.		

	4. System validates the date range and retrieves only online orders whose payment is completed and verified. 5. System displays four summary figures for the selected period: Successful Orders, Tickets Sold, Amount Before Discounts and Amount Collected. 6. System displays collections by day for Today, Last 7 Days and This Month, or by month for This Quarter and This Year. The actor may expand View Full Data Table to read the chart values. 7. System displays up to 10 paid orders per page, sorted by payment time from newest to oldest. Each row shows order/payment information, movie/showtime, ticket count, ticket value before discounts, concessions, discount and amount collected. 8. The actor reviews the history and may change the period, move between pages, open an order or export CSV through the alternative flows.
Alternative Flows:	AF1: Select a Quick Period 1. At Step 3, the actor selects Today, Last 7 Days, This Month, This Quarter or This Year. 2. System sets the period from today, six days before today, the first day of the current month, the first day of the current calendar quarter or the first day of the current year, respectively, through today. 3. System uses daily chart buckets for the first three options and monthly buckets for the last two, resets the history to page 1 and repeats Steps 4–7. AF2: Select a Custom Date Range 1. The actor enters From Date and To Date and selects View History. 2. System validates the dates, retains the current daily/monthly chart interval and reloads the summaries, chart and first history page for that range. AF3: Browse History Pages 1. At Step 7, the actor selects Next Page or Previous Page when available. 2. System keeps the selected dates and chart interval, displays the requested page with at most 10 orders, and preserves the full-period summary/chart totals. AF4: View a Paid Order’s Details 1. The actor selects an order number in the history.

	2. System displays the movie, room/showtime, creation/payment times, paid status, customer, ticket/concession values, discount, amount collected and payment method. 3. System lists the order's ticket codes, seats, ticket statuses and prices, together with concession products, quantities and subtotals when present. 4. The actor selects Back to Sales History. System restores the selected period, chart interval and history page. AF5: Export CSV 1. The actor selects Export CSV from the history. 2. System checks the same permission and date filters, then exports all matching paid online orders, including rows beyond the current page, when the total is at most 10,000. 3. System returns a UTF-8 CSV containing the reporting period, generation information and order details. AF6: No Matching Orders 1. At Step 4, no completed online order matches the selected period. 2. System displays zero in all four summary figures, zero chart buckets and an empty-history message.
Exceptions:	EF1: Invalid Date Range or Request Parameters 1. System detects an invalid date, a year outside 1900–9998, a From Date after To Date, or a range longer than 366 inclusive days. A malformed/non-positive page value or unsupported chart interval is also rejected. 2. System displays the relevant validation errors and retains the submitted dates for correction; the invalid report request is not executed. EF2: Authentication or Permission Failure 1. An unauthenticated actor is redirected to login. An actor without VIEW_ONLINE_REPORTS is denied access. 2. The same checks apply to direct history, order-detail and CSV requests. EF3: Invalid or Unavailable Order 1. A malformed/non-positive order identifier is rejected. 2. A nonexistent order, a counter-sale order, or an online order without the required completed payment/date is not returned through this function. EF4: Unsupported or Oversized Export 1. The requested export format is unsupported, or more than 10,000 paid orders match the period.

	2. System reports the error. For an oversized CSV, the actor must narrow the date range; System does not silently truncate the export. EF5: Data-Loading Failure 1. System cannot retrieve the required data. 2. System displays a system-error response instead of a successful history or misleading zero totals. The actor may retry the request.
Priority:	Medium
Frequency of Use:	Daily and on demand
Business Rules:	BR-37-01: Include only bookings with Channel = Online, Status = Paid and a recorded PaidAt. For a nonzero final amount, at least one Paid online-payment record must have matching requested/received amounts and a recorded payment time. A zero-value paid booking does not require a payment-gateway transaction. BR-37-02: Filter by booking payment date, not order-creation date or showtime date. Both selected calendar dates are included, using Vietnam local-date boundaries. Orders without a reliable PaidAt are excluded from period-based results. BR-37-03: Quick periods run through today. Last 7 Days includes today and the preceding six days; This Quarter starts on 1 January, 1 April, 1 July or 1 October as applicable; This Year starts on 1 January. BR-37-04: Today, Last 7 Days and This Month use daily chart buckets. This Quarter and This Year use monthly buckets labelled MM/YYYY. Empty buckets show zero; each bucket includes only payments inside the selected period. Custom ranges retain the currently selected chart interval. BR-37-05: Successful Orders counts each qualifying booking once. Tickets Sold counts original tickets; replacement/split tickets issued during seat exchanges do not create additional sales. Repeated payment attempts do not multiply orders or ticket values. A Sweetbox ticket is counted as one ticket record, not two seats. BR-37-06: Amount Before Discounts is the sum of original ticket prices and concession subtotals. Amount Collected is the sum of the matching bookings' stored FinalAmount. Use stored transaction values; do not reapply current prices or subtract voucher/point discounts again.

	BR-37-07: The page has exactly four summary figures: Successful Orders, Tickets Sold, Amount Before Discounts and Amount Collected. All four and the chart use the full matching set, not only the displayed page. Collections are not reduced by cash refunds or cancellation-voucher values in this function. BR-37-08: History contains paid online orders only, with at most 10 orders per page, ordered by PaidAt descending and BookingID descending for equal payment times. An out-of-range positive page is clamped to the last available page. Missing displayed numeric values are shown as 0 or 0đ. BR-37-09: A qualifying zero-value booking still contributes to Successful Orders and Tickets Sold, contributes zero to Amount Collected, and is labelled as fully covered by vouchers/points. BR-37-10: Date ranges contain at most 366 calendar days. Dates must be valid and use years 1900–9998; page numbers must be positive integers. BR-37-11: CSV exports contain all matching paid orders, subject to a 10,000-row limit, and use the same access/date rules as the screen. CSV text is escaped and protected against spreadsheet formula interpretation. All UC-37 operations are read-only.
Other Information:	Related use cases: UC-13/UC-14 provide online booking/payment data; UC-31 controls access. This history has no movie, showtime-ID, booking-ID, content-tab or payment-attempt-status filters, and no adjustment/payment-inspection sections. Traceability: /manager/online-reports → OnlineSalesReportServlet → OnlineSalesReportService → OnlineSalesReportDAO → Booking, Ticket, Booking_Concession, OnlinePayment, Showtime, Movie, Room and User. TicketExchange is read only to identify replacement tickets. View: /WEB-INF/views/manager/online-reports.jsp.
Assumptions:	Stored booking/payment times and historical ticket/concession amounts are accurate, and the reporting schema, including Booking.PaidAt, is available. Payment confirmation and changes to ticket states are handled by their existing use cases; UC-37 reads the resulting records.